

## Problem Statement — Hierarchical, Demand Forecasting with Drivers + Forecast Value Add Assessment (Styles × Sizes)

### Business framing:

A clothing manufacturer must forecast weekly demand for **3 product styles**, broken down by **size**. Long-term supply planning cares about **top-line accuracy at style level**, but operations (cutting/sewing) needs **size-level forecasts** for planning. The challenge is to produce a retail sales (POS) forecasts that are **consistent across levels** (size forecasts sum to style totals) while using real demand drivers (price, distribution, retailer inventory).

Assume it is the beginning of Jan2025 and you need to generate the optimal SKU-level forecast for the next 12 months (Jan2025-Dec2025) to send to the supply chain to manufacture goods.

### Student tasks

1. Implement baseline models:
  - Naive / Seasonal Naive
  - ETS/ARIMA (or equivalent classical time series)
  - ARIMAX / Prophet / Seasonal Linear Regression
  - ML sequence model (GRU/LSTM/etc)
  - Other advanced and hybrid approaches
2. Add **exogenous drivers** (promo flags, price, distribution changes/door count, holidays).
3. Apply **hierarchical reconciliation** (bottom-up, top-down, middle-out)
4. Evaluate by horizon (short vs mid-term) with **MAPE/WMAPE/RMSE/Bias**

### Considerations:

- How much accuracy is gained by more complex methods? Do you lose any explainability?
- What does each forecast evaluation metric tell you about the forecast?
- At what level does it make the most sense to forecast? (consider time and product aggregations)
  - How does performance vary at each level?
  - How do you reconcile the levels together?

### **Future Decisions rooted in Forecasting**

1. What input levers can you adjust to maximize demand?
2. How would you best translate retail sales into a shipment forecast for the manufacturer?
3. For the next 2 decisions: Use the following assumptions:
  - All styles share the same materials and same capacity.
  - Materials are purchased for these products 8-12 months in advance.
  - Capacity for these products is set 5-7 months in advance
  - Production/operations (cutting and sewing the garment) is set 2-3 months in advance.
- Forecast Error Near Term versus Long Term:
  - What forecasting level and method makes sense to set materials purchasing to? Capacity? Production? Should they all you one signal? Different signals? Why or why not?
- Over/Under Capacity Utilization:
  - Assume capacity is fixed at 90% of the forecast; what production trade-offs would you make to satisfy the most demand?
  - Assume capacity is fixed at 110% of the forecast total; what products would you carry extra inventory to minimize cost of holding inventory and SLOB?
  - What other data would you want to consider to make these decisions?